

Khang Tran

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Education

Princeton University

Bachelor of Arts in Physics, and minors in Engineering and Computer Science

Princeton, NJ, USA

Sept 2024 – May 2028

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Advanced Electromagnetism (PHY304), Thermal Physics (PHY301), Quantum Physics Sequence (PHY207, 208, 305), Experimental Physics (PHY312), Quantum Field Theory (PHY509, Fall 2026), Condensed Matter Theory (PHY525, Fall 2026), Honors Linear Algebra (MAT217), Multivariable Calculus (MAT201), Complex Analysis (MAT335, Fall 2026), Algorithms and Data Structures (COS226), Machine Learning and Pattern Recognition (ECE435)

NUS High School of Mathematics and Science

Diploma (High Distinction) with Honours in Engineering & Computer Science, and Majors in Mathematics, Chemistry & Physics

Singapore

Jan 2020 – Oct 2023

- **GPA:** 4.9/5.0 (NUS High, High Distinction); 4.875/5.0 (NUS dual enrollment)
- **Relevant Coursework:** Statistics, Advanced Statistics, Advanced Calculus, Advanced Mathematics; Computer Organization, Operating Systems (NUS dual enrollment)

Skills

Programming: Python, C/C++, MATLAB/Octave, Mathematica, LaTeX

Computation: HPC / SLURM multi-node pipelines, parallel computing, DFT workflows (DeepMD), PyTorch, NumPy/SciPy

Physics: Electrodynamics, Statistical Mechanics, Quantum Physics, Mechanics; electric-propulsion (Hall-effect thruster) hardware and test instrumentation

Language: English (IELTS 8.5/9.0), Vietnamese (Native)

Experience

Undergraduate Researcher

Roberto Car Group, Princeton University

Princeton, NJ, USA

June 2026 – present

- Built an auto-relaunching HPC pipeline (SLURM, multi-node) that runs **hybrid-functional DFT path-integral molecular dynamics** (CP2K, 192 atoms, 8 ring-polymer beads) to generate the training data for the group's machine-learned interatomic potentials (DeepMD), automatically recovering from job timeouts and failures.

Space Systems Engineer Intern

Aliena

Singapore

Oct 2023 – Mar 2024

Electric-propulsion company – Hall-effect thrusters for small satellites

- Designed and built a **hardware-in-the-loop (HIL) tester** for a **Hall-effect thruster's** electronic control unit (ECU) – a programmable board that impersonates the thruster, feeding the ECU the electrical signals a live unit returns under both nominal operation and failure modes – so the ECU's control firmware could be exercised and validated with no thruster in the loop, since staging real discharge faults on hardware is expensive and hazardous.
- Built an end-to-end automation pipeline for the lab's **5A hollow-cathode ignition tests**, unifying instruments that each spoke a different protocol into one common interface and exposing them through a Grafana dashboard from which researchers start tests, set variables, and monitor discharge runs hands-off.

Projects

Failure-Mode Analysis of ML Interatomic Potentials on an Ab-Initio Reference Surface

Independent study; poster, Princeton AI Fluency Lab (May 2026)

Princeton, NJ, USA

May 2026

- Compared the H_3 potential-energy surface for the $\text{H} + \text{H}_2 \rightarrow \text{H}_2 + \text{H}$ exchange reaction, computed by six pretrained machine-learned interatomic potentials (MACE-

MP-0, CHGNet, M3GNet, UMA, ORB, SevenNet) run as ASE calculators, against the BKMP2 *ab initio* reference surface — spanning **5.1–72.4 kcal/mol** MAE.

- Showed that all six models predict an incorrect transition state, and that only two recover the correct minimum-energy path even when supplied the true transition state, localizing where learned potentials fail to reproduce the underlying physics.

Weaker Arctic Amplification during CO2 Removal from a Future Warmer State
High Meadows Environmental Institute

Princeton, NJ, USA
June 2025 – Aug 2025

- Studied the effects of varying forcing strengths on the Arctic Amplification phenomenon under the guidance of Ivan Mitevski in Gabriel Vecchi's lab.

Shielding-Free Signal Noise Suppression in Portable Low-Field MRI

Singapore University of Technology and Design (SUTD)

Singapore
Apr 2022 – Aug 2023

- Developed a shielding-free signal noise suppression system for SUTD's portable magnetic resonance imager under the mentorship of Professor Huang Shaoying and her team.
- **Silver** at Singapore Science and Engineering Fair 2023
- **Poster presentation** at [International Conference for Magnetic Resonance Microscopy 2023](#)

Leadership and Teaching

Problem Setter & Category Organizer

Princeton University Science Olympiad

Princeton, NJ, USA
Sept 2025 – May 2026

- Co-set the problems for a competition category (with one other setter) and co-ran the category (3-person team) at Princeton's annual Science Olympiad invitational tournament.

Physics Olympiad Problem Setter

Singapore Physics League (SPhL) Organizing Team

Singapore
Sept 2023 – May 2025

- Set and reviewed problems for the SPhL 2024 & 2025, a national physics olympiad.

Undergraduate Physics Tutor

Princeton University Physics Department

Princeton, NJ, USA
Sept 2025 – May 2026

- Tutored undergraduates in physics through the department's peer tutoring program.

Computer Science Lab Teaching Assistant

Princeton University

Princeton, NJ, USA
Jan 2026 – May 2026

- Lab TA for COS217 (Introduction to Programming Systems) and COS226 (Algorithms & Data Structures); supported students during lab sessions with debugging and concept review.

Extracurricular Activities

Clubs: Princeton Students in Quantum, CV & NLP Reading Group, Competitive Programming, Energy Association SPARK Journal

Awards: Pyka Prize for Physics Excellence, Princeton Physics Dept. (2026), IBM Qiskit Hackathon (Champion – Hardware), COSCON 2025 1st (Undergrad), COSCON 2024 5th (Undergrad), SPhL Gold (2022 - 8th, 2023 - 3rd), Singapore Physics Olympiad Gold (2022), Singapore Junior Physics Olympiad Gold (2021)

Events: YQuantum 2025, MIT IQuHACK 2025, SpaceVision 2024